cost of requiring more timesteps to cover the given time interval. A more efficient means to get better accuracy is to use a higher order method. Higher order methods are ones that satisfy a global error estimate (for finite time intervals) of the form

$$\bar{e} \approx C(\tau)h^r,$$

where $r > 1$.

Example 2.1 The Taylor series expansion of the solution may be written $z(t+h) = z(t) + h\dot{z}(t) + (h^2/2)\ddot{z}(t) + \ldots$. Whereas a first order truncation of this series leads to Euler's method, retaining terms through second order leads to

$$z_{n+1} = z_n + h\dot{z}_n + \frac{h^2}{2}\ddot{z}_n,$$

which is referred to as the 2nd order Taylor series method. In this formula $\dot{z}_n = \boldsymbol{f}(z_n)$, and the second derivative is obtained by differentiating the differential equation itself:

$$\ddot{z}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\dot{z}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{f}(z(t)) = \boldsymbol{f}'(z(t))\dot{z}(t) = \boldsymbol{f}'(z(t))\boldsymbol{f}(z(t)),$$

so one may write the 2nd order Taylor series method as

$$z_{n+1} = z_n + h\boldsymbol{f}(z_n) + \frac{h^2}{2}\boldsymbol{f}'(z_n)\boldsymbol{f}(z_n).$$

This method generates the flow map approximation

$$\mathscr{G}_h(z) = z + h\boldsymbol{f}(z) + \frac{h^2}{2}\boldsymbol{f}'(z)\boldsymbol{f}(z).$$

Note that by the notation $\boldsymbol{f}'(z)$ where $z \in \mathbb{R}^m$ and $\boldsymbol{f} : \mathbb{R}^m \to \mathbb{R}^m$, is meant the $m \times m$ Jacobian matrix whose ij-component is $(\boldsymbol{f}'(z))_{ij} = \partial f_i/\partial z_j$. An alternative notation for $\boldsymbol{f}'$ is $\partial \boldsymbol{f}/\partial z$.

Methods like the Taylor series method offer the prospect of better accuracy in the local approximation, and smaller global error in a given simulation, but they do not necessarily resolve the more important issue relevant for very long time integrations (which we will need in molecular dynamics): the unlimited growth of perturbations from the energy surface. In molecular dynamics, we have already seen that the Euler method has growing energy error which suggests that it will be a poor scheme where the goal is to approximate the behavior of a constant energy trajectory. This same qualitative behavior is seen in other numerical methods, such as the Taylor series method. However, there are alternatives that give both higher order of accuracy and, typically, improved energy accuracy. We discuss one of the most popular schemes of this type in the next section.